ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SPL-I Final Project Report

SWE 4304: Software Project Lab-1

GROUP-15

TITLE OF THE PROJECT

CINE++ — A File-Based Cinema Hall Management System

NAME OF SUPERVISOR

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LIST OF TEAM MEMBERS

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PROJECT OVERVIEW

CINE++ is a console-based Cinema Hall Management System designed to simulate core movie theatre operations through structured programming and efficient file-handling techniques. The system manages movies, showtimes, hall layouts, bookings, pricing and ticket generation without relying on a database or graphical interface.

The project focuses on creating a modular, maintainable, and fully functional system using C++ and file-based storage. All data including users, movies, halls, seats, tickets, and transaction logs — is stored and processed using text and binary files. The goal is to create a smooth and reliable workflow similar to real cinema ticketing systems while staying within the academic constraints of SPL-1.

PROBLEM STATEMENT

Cinema hall management involves multiple interconnected tasks such as user authentication, movie listing, showtime scheduling, seat selection, ticket generation, pricing, and record keeping. In manual or poorly integrated systems, these processes become inefficient, error-prone and difficult to maintain, leading to issues like duplicate bookings, poor seat tracking and weak data organization.

In this project, the challenge was to address these issues within a fully file-based system without using databases, external frameworks or a graphical interface. The system needed to ensure data persistence, secure user authentication, accurate seat management and a smooth booking experience through a console application.

Thus, the goal was to develop a complete and practical cinema hall management system under SPL-I constraints while keeping it modular, clear and extendable.

POTENTIAL SOLUTION

To address the problem, we developed a modular C++ application where each responsibility is handled by a separate class or service. The system begins with authentication and directs users to either the admin or user menu based on their role, while data is stored using text files through a central file manager.

The solution includes:
A main controller for login, signup, and guest access; separate admin and user services; model classes for users, movies, showtimes, and seat maps; and file-based storage for all system data. Additional helper services handle pricing, ticketing, reviews, and snack orders.

This structure keeps the system organized and ensures smooth coordination between components. **For example**, seat booking involves multiple services working together. The modular design also makes the system easier to debug, maintain and extend.

IMPLEMENTED FEATURES

1. User Authentication and Access Control

<u>The system includes:</u>

1) Signup and login functionality
2) Password validation
3) Hidden password input while typing
4) Hashed password storage using built-in C++ hashing (`std::hash<string>`)

5) Role-based access for admin and regular users
6) Guest login using Bangladesh phone number validation
7) Automatic guest numbering through a guest counter file

This improves both usability and data safety compared to storing plain-text credentials.

2. Admin Functionalities

The admin module supports:

1) Adding movies
2) Deleting movies
3) Viewing all movies
4) Adding showtimes
5) Viewing showtimes
6) Deleting showtimes
7) Viewing seat maps for specific showtimes

Showtime creation includes validation for hall number and date range. The system uses predefined time slots and binds each show to a specific hall.

3. Movie Browsing and Discovery

The user side includes several browsing improvements beyond the original proposal:

1) Viewing all movies
2) Viewing detailed movie information
3) Showing current showtimes for each movie
4) Trending movie listing based on average ratings
5) Filtering movies by genre and language

6) Searching movie names by keyword

These additions make the system more user-friendly and improve discoverability of content.

4. Hall and Seat Management

The project supports multiple halls with separate seat layouts. Each hall has a predefined size:

Hall 1: large
Hall 2: medium
Hall 3: small

For every hall and showtime combination, the system maintains a separate seat map file. This means each show has its own persistent seat layout. Seats are represented in a two-dimensional grid and are marked as:

O for **available**
X for **booked**

<u>The system supports:</u>

1) Seat layout display
2) Seat availability checking
3) Booking selected seats
4) Releasing seats on cancellation
5) Saving updated seat states into files

5. Smart Seat Recommendation

One of the improved parts of the final project is the seat recommendation logic. Instead of simply choosing arbitrary seats, the system recommends seats based on closeness to the center of the hall.

Implemented recommendation features:

1) Best single seat suggestion
2) Best group seat suggestion
3) Preference for center-area seats
4) Contiguous group seat recommendation where possible

This made the booking experience much more realistic and polished.

6. Booking and Pricing System

The booking system supports:

1) Showtime selection
2) Multi-seat input
3) Duplicate seat input prevention
4) Validation of seat availability
5) Booking confirmation through ticket generation

The pricing system includes:

1) Fixed base ticket price
2) Group discount logic based on the number of seats booked
3) Optional global discount support through file-based configuration
4) Price breakdown before final ticket generation

This module significantly improved from the initial scope by making the pricing workflow clearer and more reusable.

7. Ticket Generation, Cancellation and Refund

The ticketing module supports:

1) Unique ticket ID generation
2) Formatted printable ticket output
3) Ticket saving to file
4) Ticket existence checking
5) Cancellation by ticket ID
6) Automatic seat release after cancellation
7) Refund calculation based on remaining time before show
8) refund logging
9) Cancelled ticket summary printing

Refund rules in the final implementation are time-sensitive:

1) **80%** refund if cancelled **at least 6 hours** before show
2) **50%** refund if cancelled **at least 2 hours** before show
3) **20%** refund if cancelled **at least 30 minutes** before show
4) Cancellation not allowed **if less than 30 minutes** remain or **the show has started**

8. Ratings and Reviews

The project implements both ratings and written reviews.

Rating features:

1) Add or update rating
2) Average rating calculation per movie
3) Total rating count display
4) Use of ratings in trending movie selection

Review features:

1) Show all reviews for a movie
2) Add or update a review
3) Delete own review
4) Admin can delete any review

This is one of the areas that became more complete in the final project compared to the earlier proposal.

9. Snack Shop Module

A major addition in the final system is the snack ordering feature after ticket booking. Users can optionally order food items such as burgers, sandwiches, popcorn and cola. The module calculates:

1) subtotal
2) VAT
3) final snack bill

This feature makes the project more realistic and adds value beyond a basic booking system.

10. User-Friendliness and Reliability Improvements

The final project also improved the console experience through:

1) Colored terminal output
2) Structured menus
3) Input validation
4) Safer parsing of stored files
5) Graceful handling of corrupted or invalid records
6) Better browsing flow through separate menus

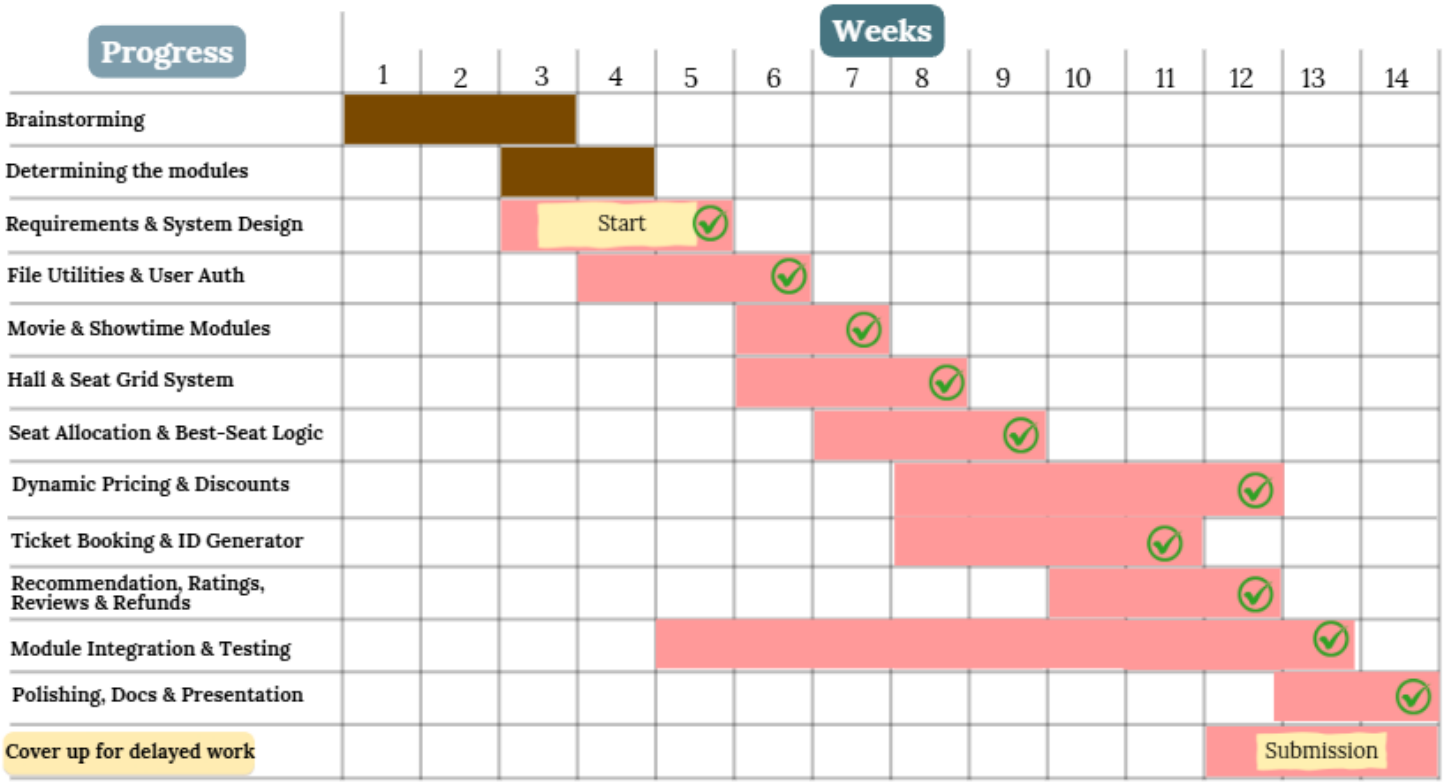
These improvements make the system more polished and presentation-ready.

TOOLS AND TECHNOLOGY

Category	Details
Programming Language	C++
Development Style	OOP, Modular Design, Menu-driven Console
Data Storage	Text files, `fstream`, `stringstream`
Libraries Used	vector, string, pair, set, map, hash, ctime, iomanip, algorithm
Security	Password hashing, input validation, safe parsing
Version Control	Git, GitHub
Environment	Console-based, No GUI, No database

PROJECT TIMELINE

The project followed a structured semester-based progression:



APPLICATION DOMAIN

The application domain of this project is cinema and entertainment management, specifically:

1) Movie show scheduling
2) Hall seat management
3) Ticket booking and cancellation
4) Customer interaction through ratings and reviews
5) Concession sales through snack ordering

The system models a simplified but practical version of a real cinema booking workflow. This makes it both academically valuable and relatable to real-world software problems.

FUTURE WORK

Although the current system is functional and feature-rich for SPL-I, several future improvements are possible:

- Automatic resetting or archiving of seat maps after shows
- Stronger password security with advanced hashing techniques
- Personalized movie recommendations based on user activity
- Admin dashboard for sales and revenue analytics
- Movie request and approval system
- Improved validation for expired showtimes
- Dynamic pricing based on time and demand
- Enhanced snack order tracking linked to tickets
- Graphical user interface for better user experience
- Migration from text files to a database-backed system

CONTRIBUTION

- **Faiyaz Tahmid (230042107)**

Contribution Area	Details
Seat Map System	Implemented complete seat map structure, layout design and visualization logic
Best Seat Algorithm (Partial)	Contributed to optimized seat recommendation system
Ticket Generation & Printing	Designed ticket generation and formatted printing system
Pricing System	Developed dynamic pricing mechanisms
Discount & Billing Logic	Implemented discount rules, promo offers, and billing calculations
Refund System	Designed refund processing logic with proper validation

Ticket Cancellation (Partial)	Contributed to cancellation workflow and validation
Ratings & Reviews (Partial)	Implemented user feedback storage and handling
Snack Shop Module (Partial)	Developed snack purchase system and billing features

- **Abrar Faiyaz (230042108)**

Contribution Area	Details
User Authentication System	Designed and implemented full login, signup, and credential validation
User/Admin/Guest Management	Managed role-based access for user, admin, and guest login
Movie Code & Details (Partial)	Contributed to structured movie data handling
Search System (Partial)	Implemented keyword-based search functionality
Best Seat Algorithm (Partial)	Assisted in optimized seat allocation logic
Ticket Generation (Partial)	Contributed to ticket creation and booking workflow
Refund System (Partial)	Contributed to refund processing and validation
Discount & Billing Logic (Partial)	Contributed to Implement discount rules, promo offers and billing calculations

- **MD. Zahin Uddin Nirob (230042160)**

Contribution Area	Details
Movie Management System	Implemented movie code, details and overall movie data management
Filtering & Search System	Developed filtering and keyword-based search functionality
Showtime Management	Designed and implemented adding, deleting, and managing showtimes
Ratings & Reviews System	Developed full user feedback system
Snack Shop Module	Implemented snack purchasing system and billing
Utility Files	Developed reusable helper functions supporting the full project
Best Seat Algorithm (Partial)	Assisted in seat optimization logic
Ticket Cancellation (Partial)	Contributed to cancellation workflow
User Authentication (Partial)	Assisted in login and authentication features
Guest Login (Partial)	Contributed to guest access system
Refund System (Partial)	Contributed to refund processing and validation
Refund System (Partial)	Contributed to refund processing logic

Overall Contribution

Area	Details
System Integration	Combined all modules into a fully functional system
Testing & Debugging	Identified, tested and fixed errors
Documentation	Prepared proposal report, final report and presentation

SOURCE REPOSITORY LINK

GitHub Repository: CINE++

CONCLUSION

CINE++ successfully demonstrates how a realistic cinema hall management workflow can be built using standard C++ and file-based persistence. The final project goes beyond a basic booking simulator by combining authentication, modular data handling, seat map management, recommendation logic, ticketing, refund policies, ratings, reviews and snack ordering into one connected system.

The project helped us apply object-oriented design, file handling, input validation, algorithmic thinking, and integration of multiple modules in a practical domain. It also showed the importance of iterative improvement because the final version became stronger than the initial proposal through better structure, better usability and additional features.

Overall, this project served as a strong practical learning experience in software design and implementation under academic constraints.